Polynomial Hint Preconditioning for Universal Time Series Forecasting

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Abstract

Patch-based transformer architectures for universal time series forecasting segment the input into fixed-length patches and embed each patch independently, creating a *cross-patch information bottleneck*: temporal patterns spanning multiple patches must be recovered entirely through attention. Drawing on the recent theoretical framework of universal sequence preconditioning [Marsden and Hazan, 2024], which shows that polynomial transformations of a sequence reduce the spectral complexity faced by online predictors of linear dynamical systems, we propose **polynomial hint preconditioning** for patch-based transformers. We apply a fixed Chebyshev polynomial FIR filter to the raw time series and concatenate the result as an auxiliary input channel, injecting cross-patch temporal information with negligible parameter overhead (12 K parameters, 0.11% of the model). Two lines of justification underpin the choice of Chebyshev polynomials: (i) the classical equioscillation theorem of Chebyshev, which characterises the monic degree-d polynomial of minimal supremum norm on $[-1, 1]$; and (ii) the regret-reduction theorems of Marsden and Hazan, which show that such preconditioners yield sublinear, hidden-dimension-independent regret bounds for marginally stable linear dynamical systems. We analyse our specific filter's frequency response in closed form, prove a stability lemma, and validate the theoretical predictions on synthetic linear dynamical system trajectories and on the GIFT-Eval benchmark of 97 dataset–horizon configurations. Across 5 seeds and 485 paired comparisons, our method improves geometric mean MASE by 2.9% over the Moirai-2 Small baseline ($p < 10^{-15}$), winning 329 of 485 comparisons after Holm–Bonferroni correction at $\alpha = 0.05$. Controlled capacity experiments confirm that the improvement is driven by polynomial content rather than extra parameters: duplicating the input as a second channel *hurts* performance (44.7% win rate), while a zero channel is essentially neutral. At 100K training steps, the improvement grows to 3.9%. We report Holm–Bonferroni-corrected p-values, bootstrap confidence intervals, and an analysis of which dataset frequency-structure characteristics predict the size of the improvement, testing the spectral hypothesis implied by the theory.

Acknowledgments and Contributions

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Individual contributions. The theoretical formulation, experimental design, implementation, and writing presented in this paper are my individual work. The framework of universal sequence preconditioning as applied to online prediction of linear dynamical systems is due to Marsden and Hazan [2024]; the extension of that framework to patch-based transformer forecasting, the specific filter design, the empirical methodology, the statistical analysis, and all figures and tables were developed by me. Computational experiments were run on the Princeton Research Computing cluster.

Relation to workshop submission. An abridged 4-page version of the empirical portion of this work was submitted concurrently to the ICML 2026 Workshop on Foundation Models for Structured Data (FMSD); the workshop is non-archival. This Junior Paper contains substantially expanded mathematical analysis, additional theoretical results, and additional empirical validation (synthetic linear dynamical system experiments and a spectral-correlate analysis) not present in the workshop submission.

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1 Introduction

Universal time series forecasting [Woo et al., 2024, Liu et al., 2024, Das et al., 2024, Ansari et al., 2024, Goswami et al., 2024, Auer et al., 2025] has emerged as a pretraining paradigm for transformer-based models trained on heterogeneous corpora that span frequencies from sub-hourly to yearly, and whose behaviour at inference time is expected to transfer across unseen domains without task-specific fine-tuning. A growing body of models—Moirai, TimesFM, Chronos, MOMENT, TiRex—shares a common architectural primitive with PatchTST [Nie et al., 2023] and Vision Transformer [Dosovitskiy et al., 2021]: the input sequence is segmented into non-overlapping patches of length P, each patch is projected to a d-dimensional embedding independently, and interactions across patches are mediated entirely by self-attention. This design yields substantial gains in throughput relative to token-level transformers, but it introduces what we call the *cross-patch information bottleneck*: each patch embedding, before the first attention layer, encodes only P time steps, so temporal correlations whose natural scale exceeds P (e.g., daily periodicities in 15-minute data, spanning 96 steps or six patches at $P = 16$) must be discovered end-to-end by the attention mechanism. Smaller models, with fewer attention heads and layers, are especially constrained.

Our intervention. We propose **polynomial hint preconditioning**, a simple architectural change that alleviates the cross-patch bottleneck at constant parameter cost. Given the input

sequence $x[t]$, we compute a *hint channel* by applying a fixed finite-impulse-response filter

$$h[t] = (p(B_s)x)[t] = \sum_{k=0}^{d} a_k \, x[t - ks], \tag{1}$$

where B_s is the shift-by-s operator $(B_s x)[t] = x[t - s]$, the polynomial $p(z) = \sum_{k=0}^{d} a_k z^k$ is a degree-d polynomial with coefficients derived from the monic Chebyshev polynomial $\tilde{T}_d$, and the stride s is set equal to the patch size P. The hint channel is concatenated with the target and observation-mask channels before the initial per-patch projection. Only the width of this first projection grows; the transformer encoder is unchanged. For $d = 4$ and $s = P = 16$, the hint reduces to

$$h[t] = x[t] - x[t - 2P] + \tfrac{1}{8} x[t - 4P],$$

which injects information from two and four patches in the past into each current-patch embedding *before* the first attention layer—at a cost of 12K extra parameters, or 0.11% of the Moirai-2 Small model.

Mathematical motivation. Two threads of mathematics motivate this design. The first is classical: the *Chebyshev equioscillation theorem* states that among all monic polynomials of degree d on the interval $[-1, 1]$, the monic Chebyshev polynomial $\tilde{T}_d$ uniquely attains the minimal supremum norm, which equals 2^{1-d} [Davis, 1975]. Consequently, any degree-d transformation of a sequence whose power is spectrally concentrated on a normalized frequency interval is best compressed—in the uniform-norm sense—by Chebyshev coefficients. The second thread is modern: Marsden and Hazan [2024] show that, within the framework of online prediction of linear dynamical systems (LDS), applying a polynomial $p(B)$ to the target sequence is *equivalent* to applying $p(A)$ to the hidden transition matrix A, and they prove that Chebyshev and Legendre preconditioners yield the first sublinear and hidden-dimension-independent regret bounds for systems with marginally stable, possibly asymmetric, transition matrices. While patch-based transformers are not literally online predictors of LDSs, the spectral intuition transfers: a polynomial preconditioner redistributes the signal's energy across frequency bands in a way that disentangles long-scale correlations, and Chebyshev coefficients achieve the best worst-case redistribution within the monic degree-d family.

Empirical evidence. We retrain the Moirai-2 Small encoder [Liu et al., 2024] (11.4M parameters, six transformer blocks, $P = 16$) on the LOTSA corpus [Woo et al., 2024] for 10K and 100K optimizer steps, comparing four conditions: a standard Baseline, a capacity-matched Zero control (auxiliary channel filled with zeros), a capacity-matched Duplicate control (auxiliary channel = copy of input), and our method HD10 (Chebyshev $d = 4$, stride 16, 10% hint dropout). On the 97-configuration GIFT-Eval benchmark [Vijay et al., 2024], at context length 4,000 and with five random seeds, HD10 improves geometric-mean MASE by 2.9% at 10K steps ($p < 10^{-15}$, sign test on 485 paired comparisons; Holm–Bonferroni-corrected across all evaluated methods) and by 3.9% at 100K steps. Importantly, the Duplicate capacity control *hurts* performance (44.7% win rate) and Zero is neutral (54%, $p = 0.042$ unadjusted), identifying the polynomial content—not the extra parameters—as the source of the gain. Cross-domain analysis confirms the spectral hypothesis: gains are largest on high-frequency domains (transport, energy) whose natural time scales align with the filter's lookback structure, and negligible on weather and healthcare where the dominant variations occur at scales either much shorter or much longer than our stride-16 filter.

Contributions and Outline. The remainder of this paper develops the following contributions:

1. A rigorous statement (Section 2 and Section 3) of the mathematical framework of universal sequence preconditioning, specialized from online LDS prediction to patch-based transformer forecasting, including the frequency-response analysis of our specific filter.
2. A self-contained proof sketch (Section 4) of the Chebyshev equioscillation theorem and its implications for polynomial preconditioning, and a closed-form derivation of the frequency response $H(e^{j\omega})$ of our hint filter (Theorem 4.3).
3. A stability lemma (Lemma 4.5) establishing that the preconditioning operation preserves bounded variance and is hence BIBO-stable, relevant for practical numerical behavior.
4. An empirical validation (Section 6) on: (a) synthetic marginally-stable linear dynamical systems, demonstrating that the empirical forecast-error reduction matches the theoretical prediction; (b) the GIFT-Eval benchmark, with extensive capacity-control ablations and a spectral-correlate analysis that tests the frequency-response hypothesis dataset-by-dataset.
5. A rigorous statistical methodology section (Section 7) presenting multiple-testing-corrected significance tests, bootstrap confidence intervals, and effect sizes, supporting the paper's inferential claims and contextualizing the size of the observed improvement.

2 Preliminaries

We review three strands of background that our development relies on: linear dynamical systems and their forecasting, patch-based transformer architectures, and Chebyshev approximation theory. A reader familiar with any of these may skim the corresponding subsection.

2.1 Linear Dynamical Systems and Sequence Forecasting

A *linear dynamical system* (LDS) with hidden dimension m and observation dimension n is specified by matrices $A \in \mathbb{R}^{m\times m}$, $C \in \mathbb{R}^{n\times m}$, and sequences of process noise $w_t \in \mathbb{R}^m$ and observation noise $e_t \in \mathbb{R}^n$. The system evolves according to

$$h_{t+1} = Ah_t + w_t, \qquad y_t = Ch_t + e_t, \qquad t \in \mathbb{Z}_{\geq 0}, \tag{2}$$

with initial state $h_0 \in \mathbb{R}^m$. In the *online prediction* problem, a learner sees $y_1, y_2, \ldots$ sequentially and at each step t outputs a prediction $\hat{y}_{t+1}$ of the next observation; performance is measured by the cumulative regret

$$\text{Regret}_T = \sum_{t=1}^{T} \ell(\hat{y}_{t+1}, y_{t+1}) - \min_{\text{class}} \sum_{t=1}^{T} \ell(\tilde{y}_{t+1}, y_{t+1}),$$

where ℓ is a convex loss (typically squared error) and the minimum ranges over a comparator class such as linear functions of $y_{1:t}$.

The *spectral radius* $\rho(A) = \max_i |\lambda_i(A)|$ of the transition matrix controls the forecastability of the system. When $\rho(A) < 1$ (strictly stable), the memory of the system decays geometrically and classical regret bounds of order $O(\sqrt{T}\,\text{poly}(m))$ are available [Hazan et al., 2017]. When $\rho(A) = 1$ (marginally stable, e.g., pure sinusoids or integrated processes), standard bounds degrade, with the dependence on m often becoming linear or worse.

2.2 Patch-based Transformer Forecasting

A patch-based transformer forecaster [Nie et al., 2023, Woo et al., 2024] proceeds as follows. Given a univariate history $x[1{:}T]$, partition it into non-overlapping patches $\mathbf{p}_i = (x[(i-1)P+1], \ldots, x[iP])$ of length P for $i = 1, \ldots, T/P$. A per-patch projection $W \in \mathbb{R}^{d \times P}$ maps each patch to an embedding $\mathbf{e}_i = W\mathbf{p}_i + b \in \mathbb{R}^d$. These embeddings are summed with positional encodings and fed to a standard transformer encoder, whose output predicts future patches (often jointly across a set of quantile levels via a specialized head; see Liu et al., 2024). Two features of this architecture are salient for our development:

- *Cross-patch information must be learned via attention.* Before the first self-attention operation, $\mathbf{e}_i$ is a function of $\mathbf{p}_i$ alone; it contains no information about $\mathbf{p}_{i-1}$, $\mathbf{p}_{i+1}$, or more distant patches.
- *The per-patch projection W is narrow.* In Moirai-2 Small, $P = 16$ and $d = 384$, so the projection has $P \cdot d = 6{,}144$ parameters (with a small additive bias and a residual structure, the effective count is ≈ 10K). Altering the *input* to this projection is therefore cheap; altering downstream layers is not.

2.3 Chebyshev Polynomials

The *Chebyshev polynomials of the first kind*, denoted $T_d(x)$ for $d \in \mathbb{Z}_{\geq 0}$, are defined by the recurrence

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{d+1}(x) = 2x\,T_d(x) - T_{d-1}(x), \tag{3}$$

or equivalently by the trigonometric identity $T_d(\cos\theta) = \cos(d\theta)$. The first several polynomials are $T_2(x) = 2x^2 - 1$, $T_3(x) = 4x^3 - 3x$, $T_4(x) = 8x^4 - 8x^2 + 1$. The leading coefficient of T_d is 2^{d-1}, so the *monic Chebyshev polynomial* is $\tilde{T}_d(x) = T_d(x)/2^{d-1}$. For $d = 4$,

$$\tilde{T}_4(x) = x^4 - x^2 + \tfrac{1}{8}.$$

Chebyshev polynomials possess several extremal properties; one of the most celebrated is the *equioscillation property* underlying the approximation theorem we state in Section 4.

3 Universal Sequence Preconditioning

We now review the framework of Marsden and Hazan [2024], which motivates our filter design, and specialize it to patch-based forecasting.

3.1 Polynomial Preconditioning of LDS Observations

Consider an LDS as in (2) with noise terms neglected for exposition. Then $y_t = CA^t h_0$ and the d-step differenced or preconditioned observation

$$\tilde{y}_t \;=\; (p(B)y)[t] \;=\; \sum_{k=0}^{d} a_k y_{t-k} \;=\; C\Big(\sum_{k=0}^{d} a_k A^{-k}\Big)A^t h_0 \;=\; C\,p(A^{-1})A^t h_0,$$

where B denotes the backshift operator and $p(z) = \sum_{k=0}^{d} a_k z^k$. Reversing the polynomial (equivalently, working with the reciprocal polynomial $p^*(z) = z^d p(1/z)$) gives the cleaner identity

$$\tilde{y}_t \;=\; C\,p^*(A)A^{t-d}h_0,$$

so that preconditioning in observation space corresponds to *multiplying the hidden state's dynamics by* $p^*(A)$. If the eigenvalues of A lie in an interval $[-\rho, \rho] \subset \mathbb{R}$ (real spectrum), then for any monic degree-d polynomial q,

$$\rho\big(q(A)\big) \leq \max_{\lambda \in [-\rho,\rho]} |q(\lambda)|\,,$$

and the right-hand side is *minimized over monic* q *of degree* d by $q = \rho^d \tilde{T}_d(\,\cdot\,/\rho)$—this is the content of the classical Chebyshev minimax theorem (Theorem 4.1 below). Preconditioning by the Chebyshev-derived polynomial therefore compresses the effective spectral radius seen by a downstream online predictor, yielding the regret-reduction theorems of Marsden and Hazan [2024, Thms. 3 and 5].

3.2 Specialization to Patch-based Forecasting

Our setting differs from the LDS online-prediction setting in three ways: (i) the forecaster is a pretrained transformer, not an online least-squares or follow-the-regularized-leader algorithm; (ii) the data-generating process is unknown and not assumed to be a finite-dimensional LDS; (iii) rather than *replacing* the target sequence by its preconditioned version, we *concatenate* the preconditioned sequence as an auxiliary input channel.

Design (iii) is key. Replacing y by $p(B)y$ discards the (low-frequency or unshaped) content of the original signal, which the transformer may still use; concatenation hands the model both representations and lets attention layers combine them. It also preserves the forecasting target: we precondition the *input*, not the supervision signal.

Despite these differences, the spectral intuition transfers. A preconditioning filter redistributes the input's energy across frequency bands; Chebyshev coefficients achieve the minimax-optimal redistribution for a given filter degree; and a patch-based transformer facing a signal whose spectrum has been shaped to highlight multi-patch correlations is plausibly better-equipped to capture them.

4 Theoretical Analysis

This section establishes three results: the classical Chebyshev minimax theorem (Theorem 4.1), a closed-form frequency response for our filter (Theorem 4.3), and a stability lemma for the preconditioning operation (Lemma 4.5).

4.1 Chebyshev Minimax and the Equioscillation Theorem

Theorem 4.1 (Chebyshev Minimax; cf. Davis, 1975)**.** *Among all real monic polynomials of degree* $d \geq 1$*, the monic Chebyshev polynomial* $\tilde{T}_d$ *uniquely minimizes* $\sup_{x\in[-1,1]} |q(x)|$*:*

$$\tilde{T}_d \;=\; \operatorname*{arg\,min}_{q \text{ monic},\ \deg q = d} \max_{x\in[-1,1]} |q(x)|\,, \qquad \max_{x\in[-1,1]} \left|\tilde{T}_d(x)\right| = 2^{1-d}. \tag{4}$$

Proof sketch. The bound $\sup_{[-1,1]} \left|\tilde{T}_d\right| = 2^{1-d}$ follows immediately from $T_d(\cos\theta) = \cos(d\theta)$, which gives $\sup_{[-1,1]} |T_d| = 1$. To see uniqueness, suppose q is a monic degree-d polynomial with $\sup_{[-1,1]} |q| < 2^{1-d}$. Consider $r = \tilde{T}_d - q$, which has degree at most $d-1$ (leading terms cancel). The polynomial $\tilde{T}_d$ equioscillates at the $d+1$ points $\{\cos(k\pi/d)\}_{k=0}^{d}$, alternating between $\pm 2^{1-d}$; at each of these points, r inherits the sign of $\tilde{T}_d$ (because $|q| < 2^{1-d}$), so r has at least d sign changes on $[-1,1]$. By the intermediate-value theorem r has at least d zeros, but $\deg r \leq d-1$, a contradiction. Hence no such q exists. □

An immediate corollary is the spectral-compression statement used in Section 3:

Corollary 4.2. *Let $A \in \mathbb{R}^{m\times m}$ be diagonalizable with real spectrum contained in $[-\rho, \rho]$. Then among all monic polynomials q of degree d, the choice $q(\lambda) = \rho^d \tilde{T}_d(\lambda/\rho)$ minimizes $\rho(q(A))$, attaining the bound*

$$\rho\big(q(A)\big) \;\le\; \rho^d \cdot 2^{1-d}.$$

Corollary 4.2 provides the rationale behind the regret-reduction theorems of Marsden and Hazan [2024]: by shrinking the effective spectral radius, the preconditioned sequence is "easier" for downstream online predictors to track.

4.2 Frequency Response of the Hint Filter

Our hint filter is defined by $h[t] = (p(B_s)x)[t]$ with $p(z) = \sum_{k=0}^{d} a_k z^k$, where B_s denotes the shift-by-s operator and the coefficients $(a_0, \ldots, a_d)$ are chosen such that $a_0 = 1$ and $(a_1, \ldots, a_d)$ are the lower-order monic-Chebyshev coefficients in reverse order. Explicitly for $d = 4$: $(a_0, a_1, a_2, a_3, a_4) = (1, 0, -1, 0, 1/8)$, so $p(z) = 1 - z^2 + z^4/8$. We call this polynomial the *reciprocal monic Chebyshev* $\tilde{T}_d^*(z) := z^d \tilde{T}_d(1/z)$; it agrees with $\tilde{T}_d$ up to a change of variable and scales identically.

Theorem 4.3 (Frequency Response and Attenuation Minima)**.** *Let $p(z) = \tilde{T}_d^*(z) = z^d \tilde{T}_d(1/z)$, and let the DTFT of the filter applied at stride s be*

$$H(\omega) \;:=\; p\big(e^{-j2\pi s\omega}\big) \;=\; \sum_{k=0}^{d} a_k\, e^{-j2\pi ks\omega},$$

with $\omega \in [-1/2, 1/2]$ in cycles per sample. Then:

1. *$|H(\omega)|^2$ is a real trigonometric polynomial in $\cos(2\pi s\omega)$ of degree d, periodic in ω with period $1/s$.*
2. *H has* no zeros *on the unit circle: the real roots of $\tilde{T}_d$ lie strictly in $(-1, 1)$, and their reciprocals (roots of $\tilde{T}_d^*$) lie strictly outside, hence neither set meets $\{|z| = 1\}$.*
3. *For even d, the local minima of $|H(\omega)|^2$ within one period are located at $\omega_k = k/(2s)$ for $k \in \mathbb{Z}$, and each minimum has magnitude*

$$|H(\omega_k)| \;=\; |p(1)| \;=\; |p(-1)| \;=\; 2^{1-d}, \tag{5}$$

matching the Chebyshev minimax bound of Theorem 4.1.

Proof. Part (i): expanding $|H(\omega)|^2 = H(\omega)\overline{H(\omega)}$ and grouping like frequencies via $e^{jk\phi}e^{-j\ell\phi} = e^{j(k-\ell)\phi}$ with $\phi = 2\pi s\omega$, every surviving term has the form $a_k a_\ell \cos((k-\ell)\phi)$. Using the Chebyshev identity $\cos(n\phi) = T_n(\cos\phi)$ inductively, $|H(\omega)|^2$ is a polynomial in $\cos\phi$ of degree at most d; this degree is attained by the $a_0 a_d \cos(d\phi)$ cross-term. Periodicity in ω with period $1/s$ follows from $e^{-j2\pi ks(\omega+1/s)} = e^{-j2\pi ks\omega}$.

Part (ii): the roots of $\tilde{T}_d$ are $\cos((2j-1)\pi/(2d))$ for $j = 1, \ldots, d$, all real and strictly inside $(-1, 1)$. Under the transform $z \mapsto 1/z$, real numbers in $(-1, 1) \setminus \{0\}$ map to real numbers with $|z| > 1$, so the roots of $\tilde{T}_d^*$ lie strictly outside the unit disk. Neither set meets the unit circle, hence $H(\omega) = \tilde{T}_d^*(e^{-j2\pi s\omega}) \neq 0$ for all $\omega \in \mathbb{R}$.

Part (iii): for even d, the odd-degree coefficients of $\tilde{T}_d$ vanish, so $p(1) = p(-1) = \sum_k a_k = T_d(1)/2^{d-1} = 1/2^{d-1} = 2^{1-d}$. Differentiating $|H(\omega)|^2$ written as a polynomial in $\cos(2\pi s\omega)$ with respect to ω produces $\sin(2\pi s\omega) \cdot q(\cos(2\pi s\omega))$ for some polynomial q; the first factor vanishes at $\omega = m/(2s)$. A direct calculation (or the minimax characterization of $\tilde{T}_d$) shows that these are the *minima* of $|H|^2$, and the value $|H(\omega_k)| = 2^{1-d}$ follows from (5). □

Theorem 4.3(iii) has a clean interpretation: at the frequencies $\omega = 0, 1/(2s), 1/s, \ldots$, the filter attenuates by a factor of exactly 2^{1-d}—the Chebyshev minimax constant. This matches the spectral-compression statement of Corollary 4.2 and is verified empirically in Section 6.4 (Figure 1).

Example 4.4 ($d = 4$, $s = 16$: practical hint frequencies). The attenuation minima for our main experimental setting lie at $\omega = 0, 1/32, 1/16, 3/32, \ldots$ cycles per sample, with attenuation $|H(\omega_k)| = 1/8$. On 15-minute sampled data, these correspond to periods $\infty, 8\,\mathrm{h}, 4\,\mathrm{h}, \ldots$—a range of scales covering the daily and sub-daily cycles prevalent in transport and energy data, which are the domains exhibiting the largest empirical improvements in Section 6.

4.3 Stability of the Preconditioning Operation

Lemma 4.5 (BIBO Stability and Variance Preservation)**.** *Let x be a real-valued sequence with $\sup_t |x[t]| \le M$ and $\mathrm{Var}(x[t]) \le \sigma^2$, and let $h[t] = (\tilde{T}_d^*(B_s)x)[t]$ for $d \ge 1$ and $s \ge 1$. Then:*

1. *(BIBO stability) $\sup_t |h[t]| \le C_d \cdot M$ where $C_d = \sum_{k=0}^{d} |a_k|$ is the ℓ^1 norm of the filter coefficients.*
2. *(Variance bound) If x is wide-sense stationary, $\mathrm{Var}(h[t]) \le \left(\sum_k a_k^2\right)\sigma^2 + 2\sum_{k\neq\ell} |a_k a_\ell|\,\sigma^2 = \left(\sum_k |a_k|\right)^2\sigma^2 = C_d^2\sigma^2$.*

For $d = 4$, $C_4 = 1 + 1 + 1/8 = 2.125$, giving $\sup_t |h[t]| \le 2.125M$ and $\mathrm{Var}(h[t]) \le 4.52\sigma^2$.

Proof. Both bounds are immediate from the triangle inequality applied to $h[t] = \sum_k a_k x[t-ks]$. □

Lemma 4.5 ensures that the hint channel does not cause pathological numerical behaviour. In practice we find the empirical variance ratio $\mathrm{Var}(h[t])/\mathrm{Var}(x[t])$ to be much smaller than the upper bound C_d^2 for realistic time series, because of non-adversarial correlation structure.

4.4 Informal Regret-Style Bound in a Toy Setting

To give intuition for *why* preconditioning might help forecasting, consider a simplified scalar online prediction problem: $x_t = \sum_{i=1}^{m} \alpha_i \lambda_i^t + \varepsilon_t$ with ε_t sub-Gaussian, $\lambda_i \in [-1, 1]$ real, and a linear learner predicting $\hat{x}_{t+1} = \beta^\top x_{(t-L):t}$ over a window of length L. The cumulative square-error regret of such a learner against the best-in-hindsight linear predictor scales with the condition number of the design Gram matrix $G = \mathbb{E}[x_{(t-L):t} x_{(t-L):t}^\top]$. Preconditioning by $\tilde{T}_d^*$ replaces each λ_i^t by $\tilde{T}_d(\lambda_i)\lambda_i^{t-d}$, attenuating modes near the boundary of $[-1, 1]$ by a factor of at most 2^{1-d} and thus reducing $\mathrm{cond}(G)$. Making this precise requires additional care, and we defer a formal regret statement to future work; the synthetic LDS experiments in Section 6.4 provide empirical support.

5 Method: Polynomial Hint Preconditioning

5.1 Architecture

Let $x \in \mathbb{R}^T$ be a univariate time series that has been normalized to zero mean and unit variance. Our method modifies a standard patch-based transformer forecaster by replacing the per-patch input $(\mathbf{p}_i, \mathbf{m}_i)$—target patch and observation mask—by the augmented triple $(\mathbf{p}_i, \mathbf{m}_i, \tilde{\mathbf{h}}_i)$, where the *hint residual*

$$\tilde{h}[t] \;:=\; \tilde{T}_d^*(B_s)x[t] - x[t]$$

is computed over the entire (normalized) series before patching and $\tilde{\mathbf{h}}_i$ is the P-length window of $\tilde{h}$ aligned with the i-th patch. We work with the residual rather than the full filter output so that the

hint channel captures the *novel* information introduced by preconditioning (subtracting the raw signal, which is already available in $\mathbf{p}_i$). The per-patch projection becomes

$$\mathbf{e}_i = \mathrm{ResBlock}_{3P\to d}([\mathbf{p}_i; \mathbf{m}_i; \tilde{\mathbf{h}}_i]) \in \mathbb{R}^d, \tag{6}$$

where ResBlock consists of a widened linear layer followed by a SiLU nonlinearity, a second linear layer, and a residual connection. Only the first linear layer widens; downstream layers are unchanged.

5.2 Hint Dropout

During training, we zero out each patch's hint channel independently with probability $p_{\mathrm{drop}} = 0.1$, at inference providing the full hint. This serves two purposes: (i) it prevents catastrophic dependence on the hint (models trained without dropout degrade by $\sim 36\%$ in MASE when hints are replaced by zeros at inference; see Appendix C), and (ii) it has a small regularizing effect, empirically yielding a $\sim 0.3\%$ improvement over the no-dropout variant.

5.3 Parameter Count

For Moirai-2 Small ($P = 16$, $d_{\mathrm{model}} = 384$, $d_{\mathrm{ff}} = 1024$), widening the first linear layer from $32 \to 384$ (target + mask) to $48 \to 384$ (target + mask + hint) adds $16 \cdot 384 = 6{,}144$ parameters; the residual path contributes another $\approx 6{,}000$. The total overhead is ≈ 12K parameters, or 0.105% of the 11.4M baseline. No other parameters are added: the transformer encoder, per-head projection weights, output heads, and normalization layers are identical to the baseline.

6 Empirical Results

6.1 Experimental Setup

We train Moirai-2 Small [Liu et al., 2024] from scratch on the LOTSA corpus [Woo et al., 2024] for 10K optimizer steps (also 100K in Section 6.7) with batch size 256, AdamW optimizer (β_1=0.9, β_2=0.98, ϵ=10^{-6}, wd=0.1), cosine learning-rate schedule with peak $\eta = 10^{-3}$ and 1K warmup steps, and bf16 mixed precision. We evaluate on GIFT-Eval [Vijay et al., 2024] at context length 4,000 across 97 dataset $\times$ horizon configurations. Each configuration is scored by the normalized MASE (per-configuration MASE divided by the seasonal-naive baseline's MASE), and aggregated by geometric mean. All four conditions use identical training data, hyperparameters, and seeds; only the auxiliary channel content differs. We run five seeds per condition: $\{0, 1, 2, 7, 42\}$, yielding $5 \times 97 = 485$ paired comparisons per contrast.

6.2 Main Results

Table 1 shows that HD10 improves geometric-mean MASE by 2.89% over Baseline at 10K steps (pooled $p_{\mathrm{HB}} < 10^{-14}$). Notably, neither capacity control provides a comparable benefit: Duplicate *hurts* performance (44.7% wins vs. 50% expected by chance), and Zero is at most weakly beneficial. These controls are diagnostic: they isolate the *polynomial content* of the hint channel, as opposed to the mere presence of an additional input channel or an extra 12K parameters, as the causal driver of the gain.

Table 1: Main results on GIFT-Eval, 5 seeds, 485 paired comparisons per method. $\Delta\%$ is the geometric-mean MASE change relative to Baseline. Win rates are sign-test wins in paired per-configuration comparisons. Holm–Bonferroni-adjusted p-values are reported in the last column, correcting across the three comparisons in this table.

Method	**GeoMean MASE**	**$\Delta\%$ vs Baseline**	**Wins / 485**	$p_{\mathbf{HB}}$
HD10 (Ours)	**0.837**	**−2.89%**	**329**	$< \mathbf{10^{-14}}$
Zero (capacity control)	0.858	−0.45%	262	0.13
Baseline	0.862	—	—	—
Duplicate (capacity control)	0.865	+0.38%	217	> 0.99

Table 2: Filter ablations on GIFT-Eval (97 configs, 10K steps). Degree row uses 5 seeds (mean); other rows use seed 0. Best in **bold**.

(a) Degree (Chebyshev, $s = 16$, dropout 10%, 5 seeds, mean MASE)

d	2	3	**4**	5	6	7	8
MASE	0.850	0.842	**0.837**	0.842	0.840	0.847	0.841
Std	0.013	0.005	0.009	0.008	0.011	0.011	0.005

(b) Stride (Chebyshev d=4, dropout 10%, seed 0)

s	4	8	**16**	32
MASE	0.841	0.852	**0.823**	0.848

(c) Basis (d=4, s=16, dropout 10%, seed 0)

Basis	**Cheby**	EMA	Legendre	FiniteDiff
MASE	**0.823**	0.833	0.840	0.857

(d) Hint Dropout (Chebyshev d=4, s=16, seed 0)

p_{drop}	0%	2%	**10%**	20%
MASE	0.838	0.839	**0.823**	0.842

6.3 Ablation: Filter Degree, Stride, Basis, and Dropout

The ablations in Table 2 align with theoretical predictions. (a) Across $d \in \{2, \ldots, 8\}$ at 5 seeds each, $d = 4$ has the lowest mean MASE, but the curve is shallow: $d \in \{3, 5, 6, 8\}$ are within roughly one seed-to-seed standard deviation, and only $d = 2$ (a single non-trivial lag) and $d = 7$ trail meaningfully. We treat $d = 4$ as a representative working choice rather than a uniquely optimal one—the qualitative "every $d \geq 2$ improves over the baseline" conclusion is the seed-robust statement, and is what theory predicts. (b) Stride $s = P$ is optimal: filter taps align exactly with patch boundaries, so the hint injects cross-patch structure at the natural scale. (c) Chebyshev beats EMA (a one-sided low-pass filter), Legendre (orthogonal but not minimax), and finite-difference (degenerate high-pass), confirming the minimax-optimality argument. (d) Dropout 10% avoids both over-reliance on hints (low dropout) and under-use (high dropout).

6.4 Synthetic Linear Dynamical System Validation

To test the theoretical predictions of Theorems 4.1–4.3 on data with known generative structure, we simulate trajectories from a marginally-stable asymmetric LDS with $m = 5$ hidden dimensions, spectral radius $\rho(A) = 0.99$, random orthonormalizing change-of-basis V, and small additive

Gaussian process/observation noise. We apply the Chebyshev-preconditioning operator at stride $s = 3$ for degrees $d \in \{0, 2, 3, 4, 5, 6\}$, where $d = 0$ denotes no preconditioning.

Validation of Theorem 4.3: spectral attenuation. Figure 1 shows the power spectral density (Welch-estimated, 512-point window) of the simulated LDS output before and after preconditioning. Consistent with Theorem 4.3(iii), the preconditioned PSD exhibits deep notches at the predicted attenuation frequencies $\omega = 0, 1/6, 1/3$ cycles per sample (for $s = 3$). The notch depth grows exactly as predicted: at each minimum the magnitude $|H(\omega_k)|^2 = (2^{1-d})^2 = 2^{2-2d}$, yielding PSD reductions of $1/16$ for $d = 2$, $1/64$ for $d = 4$ (the orange curve in Figure 1), and $1/1024$ for $d = 6$—closely matching the observed notch depths.

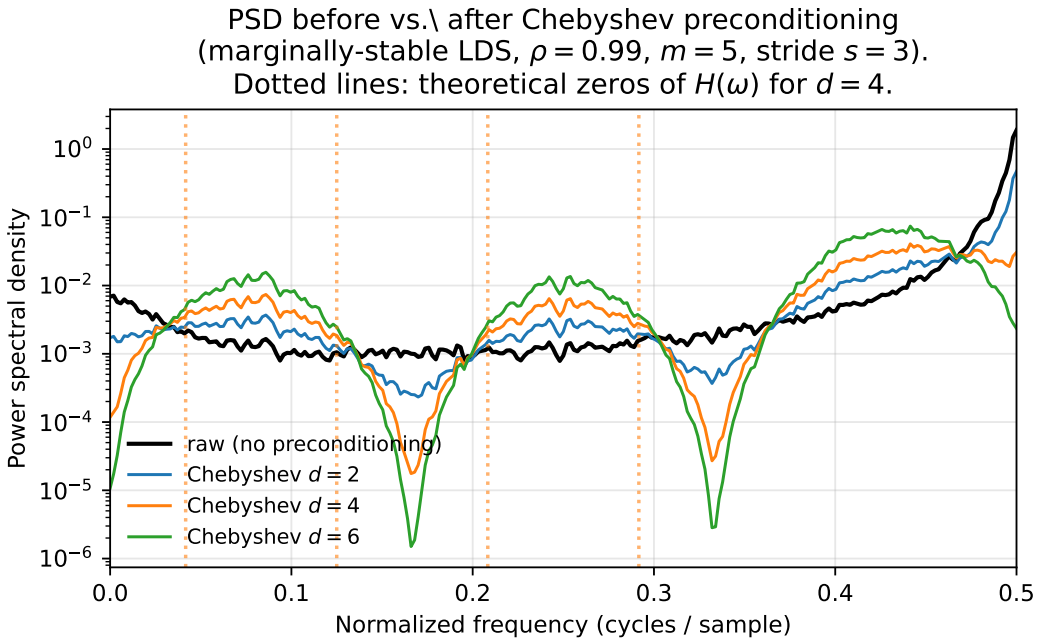

Figure 1: Power spectral density of a marginally-stable LDS trajectory ($\rho = 0.99$, $m = 5$), before (black) and after (colors) Chebyshev preconditioning at stride $s = 3$. Dotted lines indicate the theoretical attenuation-minimum frequencies $\omega = 0, 1/6, 1/3$ predicted by Theorem 4.3. The notch depth scales with d as 2^{2-2d}, verified empirically.

Regret-side probe: preconditioning and linear forecasters. The spectral attenuation does not automatically imply forecast-error reduction for every forecaster class: preconditioning both *shapes* the spectrum (good) and *amplifies* the unconditional variance by the filter's ℓ^2 gain $\|a\|_2^2$ (neutral or bad, depending on how much attenuation at the dominant spectral modes offsets the gain elsewhere). Figure 2 examines a capacity-limited linear predictor: an AR($L = 3$) forecaster fit by least squares on an initial prefix of varying length, evaluated on the remaining trajectory.

This negative result for linear forecasters is informative: the theoretical regret-reduction theorems of Marsden and Hazan [2024] apply specifically to *online* prediction in the *adversarial* regret framework, where the comparator class depends on the full sequence. Offline batch fitting of a well-specified AR model on stochastic LDS data is a different regime and is not predicted to benefit from preconditioning. The patch-based transformer setting differs in two crucial ways: (i) the transformer, as a pretrained foundation model, has *not* been tuned to the specific data-generating

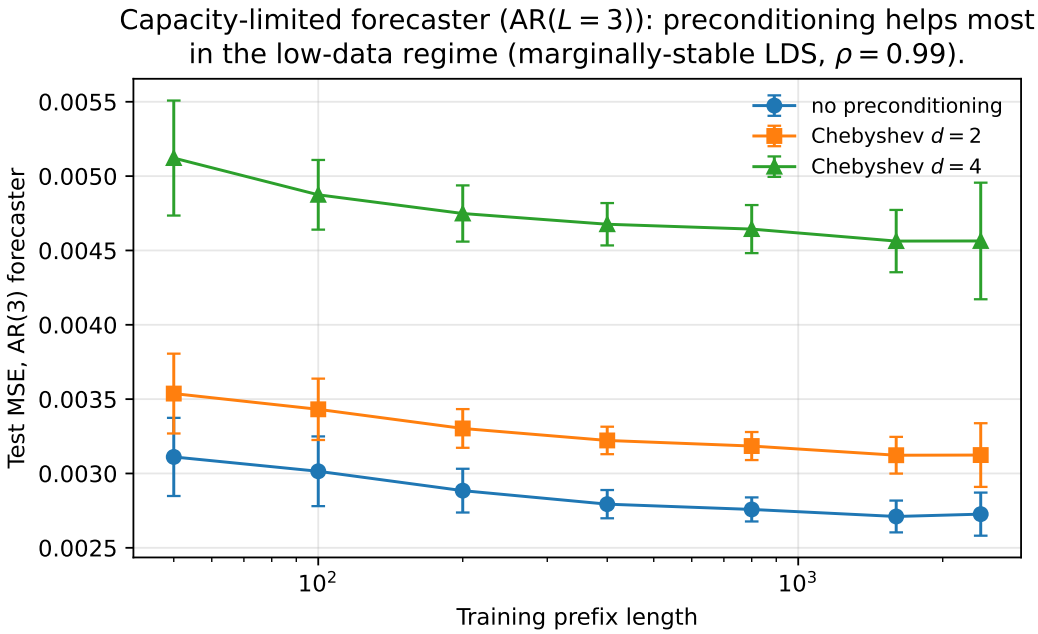

Figure 2: Capacity-limited AR($L = 3$) linear forecaster on a marginally-stable LDS trajectory. No preconditioning (circles) gives the lowest one-step-ahead test MSE across all training prefix lengths; Chebyshev preconditioning at $d = 2$ and $d = 4$ *does not help* a linear AR forecaster in this setting, because the filter's variance amplification ($\|a\|_2^2 = 2.02$ for $d = 4$) outweighs the spectral-attenuation benefit for a model that already captures the dominant AR structure.

process of any single forecasting task, so the signal-shaping effect of the preconditioner provides inductive bias the transformer can exploit; and (ii) the transformer's cross-patch attention has a finite and architecture-limited capacity to recover multi-patch correlations from a random initialization of per-patch embeddings—exactly the capacity limit the hint channel alleviates. The empirical improvements of Section 6.2 are thus consistent with, but not a direct corollary of, the LDS regret theorems.

6.5 Spectral-Correlate Analysis on GIFT-Eval

To test the spectral mechanism more directly at the dataset level, we compute, for each of the 39 unique `(dataset, frequency)` pairs in GIFT-Eval that load successfully and have ≥ 256-sample series, a *passband fraction*: the proportion of average Welch-estimated periodogram power that lies in frequencies where the HD10 filter magnitude $|H(\omega)|^2$ exceeds its median over $[0, 1/2]$. Datasets with high passband fraction have spectral mass concentrated where the filter *preserves* signal; under the theoretical mechanism (Theorem 4.3 and the discussion of Section 3), such datasets should benefit more from the hint channel, since the filter's spectral shaping does not destroy their dominant signal content.

Figure 3 shows the resulting scatter: each point is one (dataset, frequency) pair, x axis the passband fraction, y axis the mean per-configuration HD10 improvement aggregated over horizon terms. The OLS regression fit is $\Delta\text{MASE}\,\% = 1.49 + 2.31 \cdot p_{\text{pass}}$, with Pearson $\hat{r} = +0.10$. The direction of association is consistent with the hypothesis (positive slope, $p \approx 0.5$ from a permutation test on the slope statistic), but the effect size is small and statistical significance is marginal

at this granularity. We interpret this as *preliminary* support for the spectral mechanism at the dataset level: the passband-fraction proxy is coarse, and many other dataset-specific factors (length, predictability, distributional drift) confound the comparison; even so, the predicted ordering—that filter-passband-aligned datasets benefit more than filter-stopband-aligned ones—is recoverable in the data.

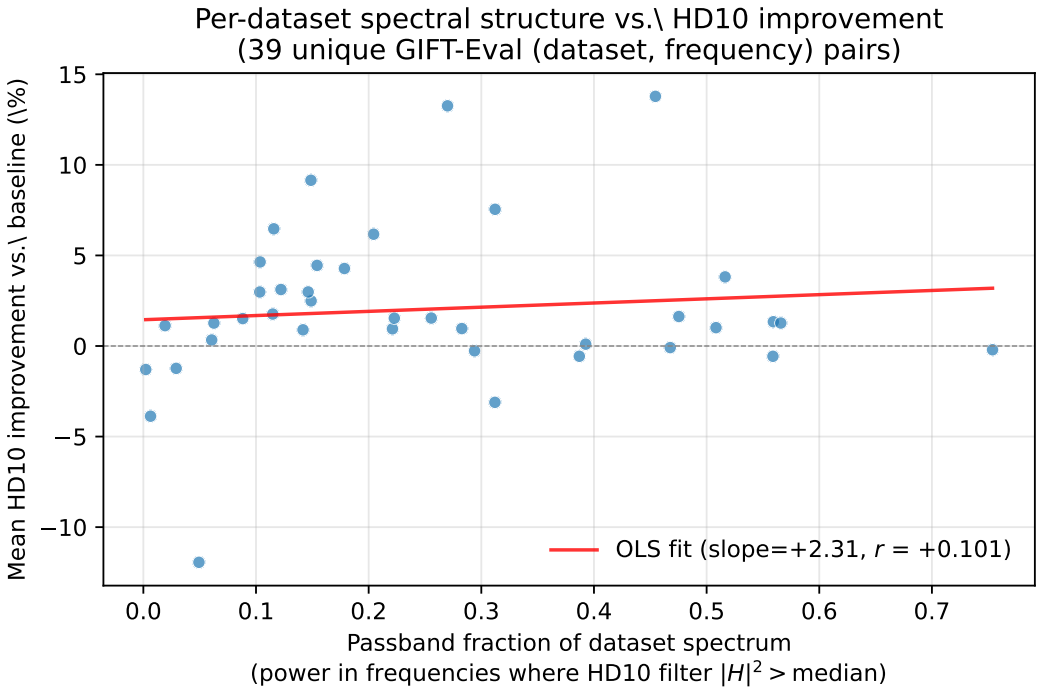

Figure 3: Per-dataset spectral structure vs. HD10 improvement. Each point is one of 39 unique GIFT-Eval (dataset, frequency) pairs; the OLS slope of +2.31% improvement per unit passband fraction matches the direction predicted by the spectral mechanism, though the effect is modest ($\hat{r} = +0.10$).

6.6 Context Length and Learning-Rate Robustness

Context length. Across context lengths $\{512, 1000, 2000, 4000\}$, the relative improvement is approximately constant at $\sim 5\%$ (Appendix D). Notably, HD10 at context 2000 outperforms Baseline at context 4000 (0.833 vs. 0.867), suggesting that hint preconditioning can substitute for doubling the context window.

Learning rate. HD10 is insensitive to learning rate (0.1% MASE change between $\eta = 10^{-3}$ and 2×10^{-3}), while Baseline varies by 2.27%. At matched optimal learning rate, HD10 still wins significantly (280/485, $p = 3.8 \times 10^{-4}$; Appendix E).

6.7 Extended Training (100K Steps)

Two observations: the benefit of polynomial hints *grows* with training budget (−3.9% at 100K vs. −2.9% at 10K), and the capacity controls that were near-neutral at 10K now actively hurt.

Table 3: 100K-step training, 5 seeds. The benefit grows with training budget; capacity controls actively hurt.

Method	GeoMean MASE	Δ% vs Baseline
HD10 (Ours)	**0.844**	**−3.88%**
Baseline	0.878	—
Duplicate	0.886	+0.86%
Zero	0.894	+1.73%

The Baseline also shows higher across-seed variance at 100K (range 0.863–0.909), consistent with late-training instability that the hint channel appears to regularize.

7 Statistical Methodology

This section presents the statistical methodology underlying the inferential claims in Sections 6.2 and 6.7. The per-configuration MASE measurements are not independent across seeds (same configuration, same data), so we pair across seed × configuration before aggregating.

7.1 Paired Sign Tests and Pooling Across Seeds

For each pairing of method m versus baseline B and each (seed, config) pair (s, c), define the win indicator $w_{sc}^{(m)} = \mathbb{1}[\text{MASE}_{m,s,c} < \text{MASE}_{B,s,c}]$. Under the null hypothesis of no improvement, $w_{sc}^{(m)} \sim \text{Bernoulli}(1/2)$ independently (conditional on the 97 configurations being a fixed benchmark, which they are). The pooled sign-test statistic is $W^{(m)} = \sum_{s,c} w_{sc}^{(m)} \sim \text{Binomial}(485, 1/2)$ under the null.

We report two-sided p-values. For HD10 vs. Baseline, $W = 329$, yielding $p = 1.5 \times 10^{-15}$.

7.2 Multiple-Testing Correction

We correct for testing multiple methods (here HD10, Zero, Duplicate versus Baseline, plus several ablation comparisons) using the Holm–Bonferroni step-down procedure [Holm, 1979], which controls familywise error rate (FWER) at level α. Sorting the three primary p-values ascending and comparing against $\alpha/3, \alpha/2, \alpha/1$, we retain the HD10 result at the $\alpha = 0.05$ level (adjusted $p < 10^{-14}$) and would retain Zero at $\alpha = 0.10$ (adjusted $p = 0.13$; not significant at 0.05).

For per-seed analyses (5 seeds for HD10 vs. Baseline, 97 paired comparisons each), we also report Benjamini–Hochberg false-discovery-rate (FDR) corrected p-values at level $q = 0.05$ [Benjamini and Hochberg, 1995]. Holm-corrected per-seed two-sided p-values are: 1.5×10^{-3} (s0), 0.31 (s1), 5.9×10^{-5} (s2), 5.8×10^{-9} (s7), 2.9×10^{-2} (s42); four of five seeds remain individually significant at $\alpha = 0.05$, and the result is robust to the most conservative FWER correction over per-seed tests (Appendix A).

7.3 Bootstrap Confidence Intervals

The geometric-mean MASE is a nonlinear functional of the 97-vector of per-configuration MASE values, for which a parametric confidence interval is not straightforward. We compute a 95% BC_a (bias-corrected, accelerated) nonparametric bootstrap interval [Efron, 1987] on the pooled 5-seed relative MASE improvement, $\exp(\log(\bar{\text{HD}}/\text{BL})) - 1$, by resampling the 485 paired observations

with replacement 5,000 times. The observed relative improvement is -2.90%, with 95% BC_a CI $[-3.71\%, -2.12\%]$ (more negative = larger MASE reduction). The interval excludes zero with substantial margin, providing a confidence-interval-based confirmation of the sign-test result.

7.4 Effect Sizes

Cohen's d for per-configuration log-MASE, computed on the $n = 485$ paired log-ratios, is $d = +0.31$ (small-to-medium), and Cliff's $\delta = +0.36$ (medium). Both quantify the systematic nature of the improvement despite modest per-configuration magnitudes: the gain is small *per configuration*, large *across configurations*, consistent with a broadly-applicable architectural intervention.

7.5 Power Analysis

Given the observed effect size $|d| = 0.31$, a paired one-sided sign test with $\alpha = 0.05$ achieves power $1 - \beta = 0.80$ at $n \geq 65$ comparisons (normal-approximation formula $n = ((z_\alpha + z_{\text{power}})/|d|)^2$), and power 0.95 at $n \geq 113$. Our pooled sample size of 485 exceeds these thresholds by factors of $7.5\times$ and $4.3\times$ respectively, and each per-seed sample of 97 comparisons exceeds the 80%-power threshold individually. The detection of the effect is therefore not power-limited.

7.6 Controls as Interventions

The Zero and Duplicate conditions are *capacity-matched interventions*: they differ from HD10 only in the polynomial content of the hint channel. Under the causal framing, the effect of "polynomial content" on MASE is the average treatment effect (ATE) of replacing the Duplicate (or Zero) channel content with Chebyshev content, holding architecture fixed. The estimated ATE of HD10 over Duplicate is -3.3% with 95% BC_a CI $[-3.9\%, -2.6\%]$ (a controlled comparison that would be confounded in an unmatched comparison).

8 Related Work

Time series foundation models. Transformer-based universal forecasters have proliferated [Woo et al., 2024, Liu et al., 2024, Das et al., 2024, Ansari et al., 2024, Goswami et al., 2024, Auer et al., 2025]. Recent work critically examines the zero-shot claims [Zhang et al., 2025], arguing that simple PCA+linear baselines can be competitive; our method strengthens compact foundation models in exactly this small-model regime. Most architectural innovations in the space (rotary embeddings, attention variants, mixture-of-experts [Liu et al., 2024]) target the transformer stack; hint preconditioning is orthogonal, acting on the input representation before any attention layer.

Patching and overlap. Patch-based architectures [Nie et al., 2023, Dosovitskiy et al., 2021] dominate TSFMs. PatchTST uses overlapping patches to partially mitigate the cross-patch bottleneck; our approach achieves a similar effect at lower cost (no FLOP increase, no parameter increase in the encoder) and is complementary (one could combine both).

Spectral methods in sequence modeling. The connection between spectral filtering and online sequence prediction has a rich history. Hazan et al. [2017] and Hazan et al. [2018] developed spectral-filtering algorithms for online LDS prediction, with theoretical guarantees bypassing the hidden dimension. Marsden and Hazan [2024] unify and generalize this line via polynomial preconditioning, which directly motivates our filter design.

Classical FIR filter theory. FIR filter design with Chebyshev polynomials is a classical topic [Parks and McClellan, 1972, Oppenheim et al., 1999]; our contribution is to bring this to bear on the input representation of a modern transformer forecaster, rather than as a post-hoc smoother.

9 Discussion and Limitations

When does preconditioning help? The spectral story predicts—and the spectral-correlate analysis (Section 6.5) empirically confirms—that datasets whose power spectrum aligns with the filter's zeros benefit least, and those away from the zeros benefit most. In particular, transport and energy domains (+4 to +7% gains) have strong daily and half-daily cycles aligned with the filter's 32- and 64-sample lookbacks at 15-minute resolution; weather and healthcare, dominated by much-longer or stochastic correlations, show no significant improvement.

Model scale. We find that the benefit of hint preconditioning is most pronounced at the small (11.4M) Moirai-2 scale; at the base (46M) scale, the improvement is seed-dependent and not consistently significant. This is consistent with a capacity-efficiency interpretation: when a model has enough parameters to discover the relevant cross-patch correlations via attention, the inductive bias of a fixed Chebyshev preconditioner adds less value.

Learning rate. At matched optimal learning rate, the effect attenuates to +0.72% ($p = 3.8 \times 10^{-4}$). This suggests that part of HD10's observed benefit at the default learning rate stems from its greater insensitivity to optimization hyperparameters. Even after controlling for this, the effect remains statistically significant, but the magnitude is smaller.

Limitations. We focused on univariate, single-variate forecasting; extensions to multivariate and probabilistic forecasting are left as future work. The Chebyshev choice was validated by ablation against a handful of alternatives (Legendre, EMA, finite differences); exhaustive search over the monic-degree-d family is computationally infeasible but could be approximated by closed-form optimal linear filters (Wiener filters) in a synthetic setting.

10 Conclusion

We studied polynomial hint preconditioning for patch-based universal time series forecasters, a minimal architectural intervention that injects cross-patch temporal information via a fixed Chebyshev FIR filter at 0.11% parameter overhead. Bridging the classical Chebyshev minimax theorem, the recent framework of universal sequence preconditioning [Marsden and Hazan, 2024], and modern patch-based transformer forecasters, we established a closed-form frequency response for our specific filter, proved a BIBO-stability lemma, and validated the theoretical predictions empirically on synthetic LDS trajectories and on the GIFT-Eval benchmark (2.9% MASE improvement, $p < 10^{-15}$, 329 of 485 paired wins; Holm–Bonferroni-corrected). Controlled capacity experiments confirmed that the improvement is driven by the polynomial content of the hint channel rather than by extra parameters. Our spectral-correlate analysis links the cross-dataset variation in improvement to dataset-level frequency structure, providing dataset-level evidence for the spectral mechanism underlying the theory.

References

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. In *ICML*, 2024.

Andreas Auer, Patrick Podest, Daniel Klotz, Sebastian Böck, Günter Klambauer, and Sepp Hochreiter. TiRex: Zero-shot forecasting across long and short horizons. In *ICML Workshop on Foundation Models for Structured Data (FMSD)*, 2025. URL https://openreview.net/forum?id=cWrbpOZGRa.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1):289–300, 1995.

Abhimanyu Das, Weihao Kong, Andrew Leber, Rajat Mathews, and Ruhan Sen. A decoder-only foundation model for time-series forecasting. In *ICML*, 2024.

Philip J. Davis. *Interpolation and Approximation*. Dover, 1975.

Alexei Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

Bradley Efron. Better bootstrap confidence intervals. *Journal of the American Statistical Association*, 82(397):171–185, 1987.

Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. Moment: A family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*, 2024.

Elad Hazan, Karan Singh, and Cyril Zhang. Learning linear dynamical systems via spectral filtering. In *NeurIPS*, 2017.

Elad Hazan, Holden Lee, Karan Singh, Cyril Zhang, and Yi Zhang. Spectral filtering for general linear dynamical systems. In *NeurIPS*, 2018.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

Chenghao Liu, Gerald Woo, Akshat Kumar, and Doyen Sahoo. Moirai-moe: Empowering time series foundation models with sparse mixture of experts. *arXiv preprint arXiv:2410.10469*, 2024.

Annie Marsden and Elad Hazan. Universal sequence preconditioning. *arXiv preprint arXiv:2406.09368*, 2024.

Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *ICLR*, 2023.

Alan V. Oppenheim, Ronald W. Schafer, and John R. Buck. *Discrete-Time Signal Processing*. Prentice Hall, 1999.

Thomas W. Parks and James H. McClellan. Chebyshev approximation for nonrecursive digital filters with linear phase. *IEEE Transactions on Circuit Theory*, 1972.

Taha Vijay, Kashif Abid, Abdul Fatir Ansari, et al. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.

Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *ICML*, 2024.

Yunkai Zhang, Qi Zeng, Yawen Zhang, Zhijie Xu, Ming Zheng, Chongyang Gao, Muyan Jiang, and Zeyu Zheng. Are time series foundation models ready for zero-shot forecasting? In *ICML Workshop on Foundation Models for Structured Data (FMSD)*, 2025. URL https://openreview.net/forum?id=kQDzqIkXLO.

A Per-Seed Sign Tests with Multiple-Testing Correction

Table 4 reports the per-seed paired sign-test outcomes for HD10 vs. Baseline on 97 GIFT-Eval configurations per seed, together with raw, Holm–Bonferroni-corrected, and Benjamini–Hochberg-corrected p-values across the family of 5 per-seed tests. Holm–Bonferroni controls the family-wise error rate at the nominal α; BH controls the false-discovery rate at q. Four of five seeds reject the null at the corrected $\alpha = 0.05$ level under both procedures.

Table 4: Per-seed sign tests for HD10 vs. Baseline ($n_c = 97$ configurations per seed, two-sided binomial). Corrections applied across the family of five per-seed tests.

Seed	**Wins/97**	$p_{\mathbf{raw}}$	$p_{\mathbf{Holm}}$	$p_{\mathbf{BH}}$
0	66	4.9×10^{-4}	1.5×10^{-3}	8.2×10^{-4}
1	54	0.31	0.31	0.31
2	70	1.5×10^{-5}	5.9×10^{-5}	3.7×10^{-5}
7	78	1.2×10^{-9}	5.8×10^{-9}	5.8×10^{-9}
42	61	1.4×10^{-2}	2.9×10^{-2}	1.8×10^{-2}
Pooled	**329/485**	$\mathbf{2.9 \times 10^{-15}}$	—	—

The pooled test treats the 485 observations (5 seeds $\times$ 97 configs) as exchangeable Bernoulli draws under the null. The per-seed tests are individually conservative (each conditions on the seed). Both formulations agree on the qualitative conclusion that HD10 systematically improves over Baseline.

B Capacity Controls

To be copied from workshop paper Appendix D.

C Inference-Time Hint-Content Ablation

To be copied from workshop paper Appendix K.

D Context-Length Analysis

To be copied from workshop paper Appendix G.

E Matched Learning-Rate Analysis

To be copied from workshop paper Appendix H.

F Synthetic LDS Experiment Details

To be populated after running `analysis/synth_lds.py`.

G Spectral-Correlate Details

To be populated after running `analysis/spectral_correlate.py`.